

## Mark Scheme

Q1.

| Question   | Scheme                                                                                              | Marks | AOs  |
|------------|-----------------------------------------------------------------------------------------------------|-------|------|
| (a)        | Attempts $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$                                                      | M1    | 1.1b |
|            | $\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} \quad (= 2\sqrt{3} \cos t)$                        | A1    | 1.1b |
|            | (2)                                                                                                 |       |      |
| (b)        | Substitutes $t = \frac{2\pi}{3}$ in $\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} = (-\sqrt{3})$ | M1    | 2.1  |
|            | Uses gradient of normal $= -\frac{1}{dy/dx} = \left(\frac{1}{\sqrt{3}}\right)$                      | M1    | 2.1  |
|            | Coordinates of $P = \left(-1, -\frac{\sqrt{3}}{2}\right)$                                           | B1    | 1.1b |
|            | Correct form of normal $y + \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}(x+1)$                           | M1    | 2.1  |
|            | Completes proof $\Rightarrow 2x - 2\sqrt{3}y - 1 = 0$ *                                             | A1*   | 1.1b |
|            | (5)                                                                                                 |       |      |
| (c)        | Substitutes $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$ into $2x - 2\sqrt{3}y - 1 = 0$                 | M1    | 3.1a |
|            | Uses the identity $\cos 2t = 2\cos^2 t - 1$ to produce a quadratic in $\cos t$                      | M1    | 3.1a |
|            | $\Rightarrow 12\cos^2 t - 4\cos t - 5 = 0$                                                          | A1    | 1.1b |
|            | Finds $\cos t = \frac{5}{6}, \frac{1}{2}$                                                           | M1    | 2.4  |
|            | Substitutes their $\cos t = \frac{5}{6}$ into $x = 2\cos t, y = \sqrt{3} \cos 2t,$                  | M1    | 1.1b |
|            | $Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$                                                | A1    | 1.1b |
|            | (6)                                                                                                 |       |      |
| (13 marks) |                                                                                                     |       |      |

**Notes:**

(a)

**M1:** Attempts  $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$  and achieves a form  $k \frac{\sin 2t}{\sin t}$  Alternatively candidates may apply the double angle identity for  $\cos 2t$  and achieve a form  $k \frac{\sin t \cos t}{\sin t}$

**A1:** Scored for a correct answer, either  $\frac{\sqrt{3} \sin 2t}{\sin t}$  or  $2\sqrt{3} \cos t$

(b)

**M1:** For substituting  $t = \frac{2\pi}{3}$  in their  $\frac{dy}{dx}$  which must be in terms of  $t$

**M1:** Uses the gradient of the normal is the negative reciprocal of the value of  $\frac{dy}{dx}$ . This may be seen in the equation of  $l$ .

**B1:** States or uses (in their tangent or normal) that  $P = \left(-1, -\frac{\sqrt{3}}{2}\right)$

**M1:** Uses their numerical value of  $-1/\frac{dy}{dx}$  with their  $\left(-1, -\frac{\sqrt{3}}{2}\right)$  to form an equation of the normal at  $P$

**A1\*:** This is a proof and all aspects need to be correct. Correct answer only  $2x - 2\sqrt{3}y - 1 = 0$

(c)

**M1:** For substituting  $x = 2\cos t$  and  $y = \sqrt{3} \cos 2t$  into  $2x - 2\sqrt{3}y - 1 = 0$  to produce an equation in  $t$ . Alternatively candidates could use  $\cos 2t = 2\cos^2 t - 1$  to set up an equation of the form  $y = Ax^2 + B$ .

**M1:** Uses the identity  $\cos 2t = 2\cos^2 t - 1$  to produce a quadratic equation in  $\cos t$

In the alternative method it is for combining their  $y = Ax^2 + B$  with  $2x - 2\sqrt{3}y - 1 = 0$  to get an equation in just one variable

**A1:** For the correct quadratic equation  $12\cos^2 t - 4\cos t - 5 = 0$

Alternatively the equations in  $x$  and  $y$  are  $3x^2 - 2x - 5 = 0$   $12\sqrt{3}y^2 + 4y - 7\sqrt{3} = 0$

**M1:** Solves the quadratic equation in  $\cos t$  (or  $x$  or  $y$ ) and rejects the value corresponding to  $P$ .

**M1:** Substitutes their  $\cos t = \frac{5}{6}$  or their  $t = \arccos\left(\frac{5}{6}\right)$  in  $x = 2\cos t$  and  $y = \sqrt{3} \cos 2t$

If a value of  $x$  or  $y$  has been found it is for finding the other coordinate.

**A1:**  $Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$ . Allow  $x = \frac{5}{3}, y = \frac{7}{18}\sqrt{3}$  but do not allow decimal equivalents.

| Question  | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks | AOs  |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| (a)       | $y = \operatorname{cosec}^3 \theta \Rightarrow \frac{dy}{d\theta} = -3 \operatorname{cosec}^2 \theta \operatorname{cosec} \theta \cot \theta$                                                                                                                                                                                                                                                                                                                             | B1    | 1.1b |
|           | $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$                                                                                                                                                                                                                                                                                                                                                                                                              | M1    | 1.1b |
|           | $\frac{dy}{dx} = \frac{-3 \operatorname{cosec}^3 \theta \cot \theta}{2 \cos 2\theta}$                                                                                                                                                                                                                                                                                                                                                                                     | A1    | 1.1b |
|           | (3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |      |
| (b)       | $y = 8 \Rightarrow \operatorname{cosec}^3 \theta = 8 \Rightarrow \sin^3 \theta = \frac{1}{8} \Rightarrow \sin \theta = \frac{1}{2}$                                                                                                                                                                                                                                                                                                                                       | M1    | 3.1a |
|           | $\theta = \frac{\pi}{6} \Rightarrow \frac{dy}{dx} = \frac{-3 \operatorname{cosec}^3 \left( \frac{\pi}{6} \right) \cot \left( \frac{\pi}{6} \right)}{2 \cos \left( \frac{2\pi}{6} \right)} = \dots$ <p style="text-align: center;">or</p> $\sin \theta = \frac{1}{2} \Rightarrow \frac{dy}{dx} = \frac{-3 \times \frac{\cos \theta}{\sin^3 \theta}}{2(1 - 2 \sin^2 \theta)} = \frac{-3 \times 8 \times \frac{\sqrt{3}/2}{1/2}}{2 \left( 1 - 2 \times \frac{1}{4} \right)}$ | M1    | 2.1  |
|           | $= -24\sqrt{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                           | A1    | 2.2a |
|           | (3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |      |
| (6 marks) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |      |

| Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(a)</p> <p>B1: Correct expression for <math>\frac{dy}{d\theta}</math> seen or implied in any form e.g. <math>\frac{-3 \cos \theta}{\sin^4 \theta}</math></p> <p>M1: Obtains <math>\frac{dx}{d\theta} = k \cos 2\theta</math> or <math>\alpha \cos^2 \theta + \beta \sin^2 \theta</math> (from product rule on <math>\sin \theta \cos \theta</math>)</p> <p>and attempts <math>\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}</math></p> <p>A1: Correct expression in any form.</p> <p>May see e.g. <math>\frac{-3 \cos \theta}{2 \sin^4 \theta \cos 2\theta}, -\frac{3}{4 \sin^4 \theta \cos \theta - 2 \sin^3 \theta \tan \theta}</math></p> <p>(b)</p> <p>M1: Recognises the need to find the value of <math>\sin \theta</math> or <math>\theta</math> when <math>y = 8</math> and uses the <math>y</math> parameter to establish its value. This should be correct work leading to <math>\sin \theta = \frac{1}{2}</math> or e.g. <math>\theta = \frac{\pi}{6}</math> or <math>30^\circ</math>.</p> <p>M1: Uses their value of <math>\sin \theta</math> or <math>\theta</math> in their <math>\frac{dy}{dx}</math> from part (a) (working in exact form) in an attempt to obtain an exact value for <math>\frac{dy}{dx}</math>. May be implied by a correct exact answer.</p> <p>If no working is shown but an exact answer is given you may need to check that this follows their <math>\frac{dy}{dx}</math>.</p> <p>A1: Deduces the correct gradient</p> |

Q3.

| Question | Scheme                                                                                                           | Marks    | AOs          |
|----------|------------------------------------------------------------------------------------------------------------------|----------|--------------|
| (a)      | Attempts $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{4\sec^2 t \tan t}{2\sec^2 t} (= 2 \tan t)$ | M1<br>A1 | 1.1b<br>1.1b |
|          | At $t = \frac{\pi}{4}$ , $\frac{dy}{dx} = 2, x = 3, y = 7$                                                       | M1       | 2.1          |
|          | Attempts equation of normal $y - 7 = -\frac{1}{2}(x - 3)$                                                        | M1       | 1.1b         |
|          | $y = -\frac{1}{2}x + \frac{17}{2}$ *                                                                             | A1*      | 2.1          |
|          | (5)                                                                                                              |          |              |
| (b)      | Attempts to use $\sec^2 t = 1 + \tan^2 t \Rightarrow \frac{y-3}{2} = 1 + \left(\frac{x-1}{2}\right)^2$           | M1       | 3.1a         |
|          | $\Rightarrow y - 3 = 2 + \frac{(x-1)^2}{2} \Rightarrow y = \frac{1}{2}(x-1)^2 + 5$ *                             | A1*      | 2.1          |
|          | (2)                                                                                                              |          |              |

|  |                                                                                                                                                                                                                                                      |    |      |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|
|  | (b) Alternative 1:                                                                                                                                                                                                                                   |    |      |
|  | $y = \frac{1}{2}(x-1)^2 + 5 = \frac{1}{2}(2 \tan t + 1 - 1)^2 + 5$<br>$= \frac{1}{2}4 \tan^2 t + 5 = 2(\sec^2 t - 1) + 5$                                                                                                                            | M1 | 3.1a |
|  | $= 2\sec^2 t + 3 = y$ *                                                                                                                                                                                                                              | A1 | 2.1  |
|  | (b) Alternative 2:                                                                                                                                                                                                                                   |    |      |
|  | $x = 2 \tan t + 1 \Rightarrow t = \tan^{-1}\left(\frac{x-1}{2}\right) \Rightarrow y = 2\sec^2\left(\tan^{-1}\left(\frac{x-1}{2}\right)\right) + 3$<br>$\Rightarrow y = 2\left(1 + \tan^2\left(\tan^{-1}\left(\frac{x-1}{2}\right)\right)\right) + 3$ | M1 | 3.1a |
|  | $\Rightarrow y = 2\left(1 + \left(\frac{x-1}{2}\right)^2\right) + 3 = \frac{1}{2}(x-1)^2 + 5$ *                                                                                                                                                      | A1 | 2.1  |
|  | (b) Alternative 3:                                                                                                                                                                                                                                   |    |      |
|  | $\frac{dy}{dx} = 2 \tan t = x - 1 \Rightarrow y = \int (x-1) dx = \frac{x^2}{2} - x + c$<br>$(3, 7) \rightarrow 7 = \frac{3^2}{2} - 3 + c \Rightarrow c = \frac{11}{2}$                                                                              | M1 | 3.1a |
|  | $\frac{x^2}{2} - x + \frac{11}{2} = \frac{1}{2}(x^2 - 2x) + \frac{11}{2} = \frac{1}{2}(x-1)^2 - \frac{1}{2} + \frac{11}{2} = \frac{1}{2}(x-1)^2 + 5$ *                                                                                               | A1 | 2.1  |

|            |                                                                                                                                                                                                                                                                                              |     |      |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|
| (c)        | Attempts the lower limit for $k$ :<br>$\frac{1}{2}(x-1)^2 + 5 = -\frac{1}{2}x + k \Rightarrow x^2 - x + (11-2k) = 0$<br>$b^2 - 4ac = 1 - 4(11-2k) = 0 \Rightarrow k = \dots$                                                                                                                 | M1  | 2.1  |
|            | $(k =) \frac{43}{8}$                                                                                                                                                                                                                                                                         | A1  | 1.1b |
|            | Attempts the upper limit for $k$ :<br>$(x, y)_{t=-\frac{\pi}{4}} : t = -\frac{\pi}{4} \Rightarrow x = 2 \tan\left(-\frac{\pi}{4}\right) + 1 = -1, y = 2 \sec^2\left(-\frac{\pi}{4}\right) + 3 = 7$<br>$(-1, 7), y = -\frac{1}{2}x + k \Rightarrow 7 = \frac{1}{2} + k \Rightarrow k = \dots$ | M1  | 2.1  |
|            | $(k =) \frac{13}{2}$                                                                                                                                                                                                                                                                         | A1  | 1.1b |
|            | $\frac{43}{8} < k \leq \frac{13}{2}$                                                                                                                                                                                                                                                         | A1  | 2.2a |
|            |                                                                                                                                                                                                                                                                                              | (5) |      |
| (12 marks) |                                                                                                                                                                                                                                                                                              |     |      |
| Notes:     |                                                                                                                                                                                                                                                                                              |     |      |

(a) **Must use parametric differentiation to score any marks in this part and not e.g. Cartesian form**

M1: For the key step of attempting  $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ . There must be some attempt to differentiate both

parameters however poor and divide the right way round so using  $\frac{dy}{dx} = \frac{y}{x}$  scores M0.

This may be implied by e.g.  $\frac{dx}{dt} = 2 \sec^2 t$ ,  $\frac{dy}{dt} = 4 \sec^2 t \tan t$ ,  $t = \frac{\pi}{4} \Rightarrow \frac{dx}{dt} = 4$ ,  $\frac{dy}{dt} = 8 \Rightarrow \frac{dy}{dx} = 2$

A1:  $\frac{dy}{dx} = \frac{4 \sec^2 t \tan t}{2 \sec^2 t}$ . Correct expression in any form. May be implied as above.

Condone the confusion with variables as long as the intention is clear e.g.

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{4 \sec^2 x \tan x}{2 \sec^2 x} (= 2 \tan x)$  and allow subsequent marks if this is interpreted correctly

M1: For attempting to find the values of  $x, y$  and the gradient at  $t = \frac{\pi}{4}$  AND getting at least two correct.

Follow through on their  $\frac{dy}{dx}$  so allow for any two of  $x = 3$ ,  $y = 7$ ,  $\frac{dy}{dx} = 2$  (or their  $\frac{dy}{dx}$  at  $t = \frac{\pi}{4}$ )

Note that the  $x = 3$ ,  $y = 7$  may be seen as e.g. (3, 7) on the diagram. There must be a non-trivial  $\frac{dy}{dx}$  for this mark e.g. they must have a  $\frac{dy}{dx}$  to substitute into.

M1: For a correct attempt at the normal equation using their  $x$  and  $y$  at  $t = \frac{\pi}{4}$  with the negative

reciprocal of their  $\frac{dy}{dx}$  at  $t = \frac{\pi}{4}$  having made some attempt at  $\frac{dy}{dx}$  and all correctly placed.

For attempts using  $y = mx + c$  they must reach as far as a value for  $c$  using their  $x$  and  $y$  at  $t = \frac{\pi}{4}$

with the negative reciprocal of their  $\frac{dy}{dx}$  at  $t = \frac{\pi}{4}$  all correctly placed.

A1\*: Proceeds with a clear argument to the given answer with no errors.



(b)

**M1:** Attempts to use  $\sec^2 t = 1 + \tan^2 t$  oe to obtain an equation involving  $y$  and  $(x-1)^2$

E.g. as above or e.g.  $y = 2\sec^2 t + 3 = 2\left(1 + \tan^2 t\right) + 3 = 2\left(1 + \left(\frac{x-1}{2}\right)^2\right) + 3$  for M1 and then

$$y = \frac{1}{2}(x-1)^2 + 5^* \text{ for A1}$$

**A1\*:** Proceeds with a clear argument to the given answer with no errors

**Alternative 1:**

**M1:** Uses the given result, substitutes for  $x$  and attempts to use  $\sec^2 t = 1 + \tan^2 t$  oe

**A1:** Proceeds with a clear argument to the  $y$  parameter and makes a (minimal) conclusion e.g. “ $= y$ ”  
QED, hence proven etc.

**Alternative 2:**

**M1:** Uses the  $x$  parameter to obtain  $t$  in terms of arctan, substitutes into  $y$  and attempts to use  
 $\sec^2 t = 1 + \tan^2 t$  oe

**A1:** Proceeds with a clear argument to the given answer with no errors

**Alternative 3:**

**M1:** Uses  $\frac{dy}{dx}$  from part (a) to express  $\frac{dy}{dx}$  in terms of  $x$ , integrates and uses (3, 7) to find “ $c$ ” to reach  
a Cartesian equation.

**A1:** Proceeds with a clear argument to the given answer with no errors

**Allow the marks for (b) to score anywhere in their solution e.g. if they find the Cartesian equation in part (a)**

(c)

**M1:** A full attempt to find the lower limit for  $k$ .

$$\frac{1}{2}(x-1)^2 + 5 = -\frac{1}{2}x + k \Rightarrow x^2 - x + (11 - 2k) = 0 \Rightarrow b^2 - 4ac = 1 - 4(11 - 2k) = 0 \Rightarrow k = \dots$$

Score **M1** for setting  $\frac{1}{2}(x-1)^2 + 5 = -\frac{1}{2}x + k$ , rearranging to 3TQ form and attempts  $b^2 - 4ac \dots 0$

e.g.  $b^2 - 4ac > 0$  or e.g.  $b^2 - 4ac < 0$  correctly to find a value for  $k$ .

**A1:**  $k = \frac{43}{8}$  oe. Look for this value e.g. may appear in an inequality e.g.  $k > \frac{43}{8}$ ,  $k < \frac{43}{8}$

**An alternative method using calculus for lower limit:**

$$y = \frac{1}{2}(x-1)^2 + 5 \Rightarrow \frac{dy}{dx} = x-1, x-1 = -\frac{1}{2} \Rightarrow x = \frac{1}{2}$$

$$x = \frac{1}{2} \Rightarrow y = \frac{1}{2}\left(\frac{1}{2}-1\right)^2 + 5 = \frac{41}{8}$$

$$y = -\frac{1}{2}x + k \Rightarrow \frac{41}{8} = -\frac{1}{4} + k \Rightarrow k = \dots$$

Score **M1** for  $\frac{dy}{dx}$  = “a linear expression in  $x$ ”, sets  $= -\frac{1}{2}$ , solves a linear equation to find  $x$  and

then substitutes into the given result in (b) to find  $y$  and then uses  $y = -\frac{1}{2}x + k$  to find a value

for  $k$ . **A1:**  $k = \frac{43}{8}$  oe. Look for this value e.g. may appear in an inequality e.g.  $k > \frac{43}{8}$ ,  $k < \frac{43}{8}$



An alternative method using parameters for lower limit:

$$y = -\frac{1}{2}x + k \Rightarrow 2\sec^2 t + 3 = -\frac{1}{2}(2\tan t + 1) + k$$

$$\Rightarrow 2(1 + \tan^2 t) + 3 = -\frac{1}{2}(2\tan t + 1) + k \Rightarrow 2\tan^2 t + \tan t + 5.5 - k = 0$$

$$b^2 - 4ac = 0 \Rightarrow 1 - 4 \times 2(5.5 - k) = 0 \Rightarrow k = \frac{43}{8}$$

Score M1 for substituting parametric form of  $x$  and  $y$  into  $y = -\frac{1}{2}x + k$ , uses  $\sec^2 t = 1 + \tan^2 t$

rearranges to 3TQ form and attempts  $b^2 - 4ac \dots 0$  or e.g.  $b^2 - 4ac > 0$  or  $b^2 - 4ac < 0$  correctly to find a value for  $k$ .

A1:  $k = \frac{43}{8}$  oe. Look for this value e.g. may appear in an inequality e.g.  $k > \frac{43}{8}$ ,  $k < \frac{43}{8}$

M1: A full attempt to find the upper limit for  $k$ . This requires an attempt to find the value of  $x$  and the value of  $y$  using  $t = -\frac{\pi}{4}$ , the substitution of these values into  $y = -\frac{1}{2}x + k$  and solves for  $k$ .

A1:  $k = \frac{13}{2}$ . Look for this value e.g. may appear in an inequality.

A1: Deduces the correct range for  $k$ :  $\frac{43}{8} < k \leq \frac{13}{2}$

Allow equivalent notation e.g.  $\left(k \leq \frac{13}{2} \text{ and } k > \frac{43}{8}\right)$ ,  $\left(k \leq \frac{13}{2} \cap k > \frac{43}{8}\right)$ ,  $\left(\frac{43}{8}, \frac{13}{2}\right]$

But not e.g.  $\left(k \leq \frac{13}{2}, k > \frac{43}{8}\right)$ ,  $\left(k \leq \frac{13}{2} \cup k > \frac{43}{8}\right)$ ,  $\left(k \leq \frac{13}{2} \text{ or } k > \frac{43}{8}\right)$  and do not allow if in terms of  $x$ .

Allow equivalent exact values for  $\frac{43}{8}$ ,  $\frac{13}{2}$

There may be other methods for finding the upper limit which are valid. If you are in any doubt if a method deserves credit then use Review.

Q4.

| Question   | Scheme                                                                                                                             | Marks | AOs  |
|------------|------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| (a)        | Uses the model to state $\frac{dV}{dt} = -c$ (for positive constant $c$ )                                                          | B1    | 3.1b |
|            | Uses $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ with their $\frac{dV}{dt} = -c$ and $\frac{dV}{dr} = 4\pi r^2$           | M1    | 2.1  |
|            | $-c = 4\pi r^2 \times \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = -\frac{c}{4\pi r^2} = -\frac{k}{r^2}$ *                            | A1*   | 2.2a |
|            |                                                                                                                                    | (3)   |      |
| (b)        | $\frac{dr}{dt} = -\frac{k}{r^2} \Rightarrow \int r^2 dr = \int -k dt$ and integrates with one side "correct"                       | M1    | 2.1  |
|            | $\frac{r^3}{3} = -kt (+\alpha)$                                                                                                    | A1    | 1.1b |
|            | Uses $t = 0, r = 40 \Rightarrow \alpha = \dots$ $\alpha = \frac{64000}{3}$                                                         | M1    | 1.1b |
|            | Uses $t = 5, r = 20$ & $\alpha = \dots \Rightarrow k = \dots$                                                                      | M1    | 3.4  |
|            | $r^3 = 64000 - 11200t$ or exact equivalent                                                                                         | A1    | 3.3  |
|            |                                                                                                                                    | (5)   |      |
| (c)        | Uses the equation of their model and proceeds to a limiting value for $t$<br>E.g. " $64000 - 11200t$ " ... $0 \Rightarrow t \dots$ | M1    | 3.4  |
|            | For times up to and including $\frac{40}{7}$ seconds                                                                               | A1ft  | 3.5b |
|            |                                                                                                                                    | (2)   |      |
| (10 marks) |                                                                                                                                    |       |      |
| Notes:     |                                                                                                                                    |       |      |





(a)

**B1:** Uses the model to state  $\frac{dV}{dt} = -c$  (for positive constant  $c$ ).

Any "letter" is acceptable here including  $k$ .

Note that  $\frac{dV}{dt} = c$  is B0 unless they state that  $c$  is a negative constant.

**M1:** For an attempt to use  $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$  with their  $\frac{dV}{dt}$  and  $\frac{dV}{dr} = 4\pi r^2$

Allow for an attempt to use  $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$  with their  $\frac{dV}{dt}$  and  $\frac{dV}{dr} = \lambda r^2$  (Any constant is fine)

There is no requirement to use the correct formula for the volume of a sphere for this mark.

**A1\*:** Proceeds to the given answer with an intermediate line equivalent to  $\frac{dr}{dt} = -\frac{c}{4\pi r^2}$

If candidate started with  $\frac{dV}{dt} = -k$  they must provide a minimal explanation how

$$\frac{dr}{dt} = -\frac{k}{4\pi r^2} \rightarrow \frac{dr}{dt} = -\frac{k}{r^2}. \text{ E.g. } \frac{1}{4\pi} \text{ is a constant so replace } \frac{k}{4\pi} \text{ with } k$$

It is not necessary to use the full formula for the volume of a sphere, eg allow  $V = \kappa r^3$  but if it has been quoted it must be correct. So using  $V = 4r^3$  can potentially score 2 of the 3 marks.

(b)

**M1:** For the key step of separating the variables correctly AND integrating one side with at least one index correct. The integral signs do not need to be seen.

**A1:** Correct integration E.g.  $\frac{r^3}{3} = -kt(+\alpha)$  or equivalent. The  $+\alpha$  is not required for this mark.

This may be awarded if  $k$  has been given a value.

**M1:** Uses the initial conditions to find a value for the constant of integration  $\alpha$

If a constant of integration is not present, or  $k$  has been given a pre defined value, then only the first two marks can be awarded in part (b)

The mark may be awarded if the equation has been adapted incorrectly. E.g. each term cube rooted.

**M1:** Uses the second set of conditions with their value of  $\alpha$  to find  $k$

This may be awarded if the equation has been adapted incorrectly. E.g. each term cube rooted.

**A1:** Obtains any correct equation for the model.

$$\text{E.g. } r^3 = 64000 - 11200t \text{ or exact equivalent such as } \frac{r^3}{3} = \frac{64000}{3} - \frac{11200}{3}t.$$

ISW after sight of a correct answer. Condone recurring decimals e.g. 21333.3 for  $\frac{64000}{3}$

Do not award if **only** the rounded/truncated decimal equivalents to say  $\frac{64000}{3}$  is used.

(c)

**M1:** Recognises that the model is only valid when  $r \geq 0$  and uses this to find  $t$ . Condone  $r > 0$

Award for an attempt to find the value of  $t$  when  $r = 0$ . See scheme.

It must be from an equation of the form  $ar^n = b - ct$ ,  $a, b, c > 0$  which give +ve values of  $t$ .

**A1ft:** Allow valid for times up to (and including)  $\frac{40}{7}$  seconds, 5.71 seconds. Allow  $t < \frac{40}{7}$  or  $t \leq \frac{40}{7}$

There is no requirement for the left hand side of the inequality, 0

States invalid for times greater than  $\frac{40}{7}$  seconds, 5.71 seconds.

Follow through on their equation so allow  $t < \text{their } \frac{64000}{11200}$  as long as this value is greater than 5

( $t = 5$  is one of the values in the question)